Lam Fung Academy · LF Academy

Primary 5 • 6th Lesson • Student Handout

Comprehensive: area + multiple digits + estimate

Cross-Topiccomprehensiveapplication • 65 minutes • 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Units 1 and 2 + Modern Education 5A Unit 1-5

Core traps:□ T1 Estimated unit error + T4 Confusion of large place values + T5 Area formula/unit trap (mixed across topics)

SSPA association: 🔴 High frequency Cross-topic comprehensive questions in sub-tests are compulsory every year, accounting for about 12-18% of Paper 1

Prerequisite knowledge: Class 1-5 (Multi-digit reading and writing and approximate values • Area formula • Unit conversion • Estimation strategy)

The goal of this class:❶ Estimating the area (estimate the size first → set the formula) ❷ Application of large numbers in area ❸ Comprehensive word questions (reading and writing + estimation + area three-in-one)

Student Name: ____________ Class: ____________ Date: ____________ Time Spent: ____________

I.Warm-Up Questions（total 5 question，5 minutes）

#	Question	Difficulty	Working Space（Show full working）
1	Read the number 2,358,400 and write the place value of "5".	Basic	
2	Estimate: How many meters is the length of a standard basketball court? Circle reasonable answers. A. 2 meters B. 20 meters C. 200 meters	Basic	
3	Calculate the area of the rectangle: length 12 cm, width 8 cm. Area = ?	Basic	
4	Fill in the blank: 5 m^2 = ______ cm^2 (Hint: 1 m^2 = 10000 cm^2)	Basic	
5	Take 3,867,452 to the "hundred thousand" and "ten thousand" places.	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Estimate the area - estimate the size first, then apply the formula 🔴 SSPA

① **Three-step method for estimating area:** Estimate size → set of formulas → take approximate values
② **Step 1 Estimate size:** Observe the shape and use known reference objects to estimate the length (for example: one step is about 1m, one floor is about 3m)
③ **Step 2 set of formulas:** Rectangular area = length × width; square area = side length × Side length
④ **Step 3 Take the approximate value:** Round the calculation result to the appropriate place value (thousands/ten thousand), and mark "approximately" when answering
⑤ **Trap T1:** The units must be consistent when estimating - you cannot directly multiply the length by m and the width by cm!

Figure 1: Estimation of football field area - first estimate the length and width, and then use the area formula

Trap detonation example (cross-topic mixed trap T1+T5)

Example 1: The picture below is the floor plan of a rectangular park. Estimate its area and express it in "10,000 m^2".

❌ **Common mistakes**

Length=200m Width=150m

→ 200+150=350

Use the perimeter formula (addition) to calculate the area (should be multiplied) or: forget to take the approximate value/wrongly write the unit

✅ **Correct solution**

$200 \times 150 = 30,000\ m^2$

$= \mathbf{30,000\ m^2}$

Step1: Length≈200m Width≈150mStep2: Area=length×width=30,000m^2Step3: 30,000=30,000

example

Example 2: A square square with side length about 85 m. Estimate its area and round your answer to the nearest thousand.

⚠ **T1 trap: The units for estimating length and width must be the same (both in m or both in cm) and cannot be mixed!**

⚠ **T5 trap: The area formula is "multiplication" not "addition"! The area of a rectangle = length × width, not length + width.**

Knowledge Point 2: Application of Large Numbers in Area - Conversion of m^2 and cm^2 🔴 SSPA Required Exam

① **Key conversion:**1 m = 100 cm → **1 m^2 = 100 × 100 = 10,000 cm^2**(not 100!)
② m^2 → cm^2 : **× 10,000**(the number will become four digits larger)
③ cm^2 → m^2 : **÷ 10,000**(the number will become four digits smaller)
④ m^2 or km^2 is commonly used for large areas (parks, cities); cm^2 is commonly used for small areas (books, desktops)
⑤ **Trap T4:**After conversion, the numbers are very large (millions, tens of millions), and it is easy to get the bit value wrong when reading and writing.

①
Remember the conversion rates
1 m^2
= 10,000 cm^2

②
m^2 → cm^2
× 10,000 numbers become larger

③
cm^2 → m^2
÷ 10,000The number becomes smaller

④
Check plausibility
The number in cm^2 must be larger than m^2

example

Example 3: The area of a playground is 3,500,000 cm^2. How many m^2 is it equal to? (write down the calculation process)

example

Example 4: The area of a house is 85 m^2. Equal to how many cm^2? Read the numerical value of the answer.

Knowledge point 3: Comprehensive application questions - multi-digit reading and writing + estimation + area calculation🔴 SSPA killer questions

① **Common comprehensive question types for sub-tests:**Test readings simultaneously in one question → Take approximate values → Calculate area with approximate values → Convert units
② **Problem-solving strategy:**Break down step by step, each sub-question is processed independently, the answer of the previous step may be the input of the next step
③ **Trap T4+T5 Mixed:**The numbers change after taking approximations, and the area formula is easy to make mistakes; converting units involves multiplication and division of large numbers
④ **Must check:**Is the unit of each answer correct? Are approximations marked "approximately"? Is the reading of large numbers correct?

Comprehensive trap example questions (T1+T4+T5 three-in-one)

Example 5: A rectangular square is about 125 m long and 85 m wide.
(a) Estimate the area of the square to the nearest thousand.
(b) Read the answer to (a).
(c) Convert the answer to (a) into cm^2.

example

Example 6: Round 2,468,000 to the 100,000th place to get A. Using A as the side length of the square (unit: m), calculate the area of the square and express it in "10,000 m^2".

🧠 **Tip: "When estimating, first estimate the length and width. Multiply the area instead of adding. m^2 becomes cm^2 multiplied by 10,000. Check the reading and writing of large numbers."**

⚠️ **Comprehensive questions are the most common mistakes: after taking the approximate value and forgetting to mark the word "approximately", one point will be deducted from the test! If the answer is calculated using approximate values, write "≈" throughout.**

III. Lesson Layered Synchronization Practice

Basic layer (total 6 questions, everyone must do)

#	Question	Difficulty	Working Space
6	Estimate the area of a rectangular playground: about 30 m long and 20 m wide. Area approx = ?	🌱	
7	4 m^2 = ______ cm^2	🌱	
8	A rectangle is 15 m long and 12 m wide. Area = ? m^2	🌱	
9	Read the number 3,050,700 and write down the values of the two "0"s in it.	🌱	
10	Take 4,567,890 to the "100,000 digit".	🌱	
11	6 m^2 = ______ cm^2. Read the answer.	🌱	

Advanced layer (total 5 questions, 🚶🚀 choose do)

#	Question	Difficulty	Working Space
12	A park is approximately 250 m long and 160 m wide. Estimated area, expressed in "10,000 m^2".	🌿	
13	The area of a playground is 4,800,000 cm^2. How many m^2 is it equal to?	🌿	
14	Compare two sites: Site A (95 m long and 65 m wide) and Site B (85 m long and 75 m wide). Which one has the larger area? How many m^2 is the difference?	🌿	
15	The area of a city park is 25,000,000 m^2. (a) Read this number (b) Express it in "square kilometers" (1 km^2 = 1,000,000 m^2)	🌿	
16	Round 4,567,890 to the "hundredthousandth digit" and use this approximate value as the side length of the square (unit: m) to estimate the area of the square.	🌿	

🌳 challenge layer (total 5 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
17	A parking lot is approximately 135 m long and 85 m wide. (a) Estimate the area to the nearest thousand (b) If each car requires 25 m^2, approximately how many cars can be parked at most?	🌳	

18	A rectangular site is 125 m long and 80 m wide. (a) Area = ? m^2 (b) Round the answer to the nearest ten thousand (c) Convert the original area to cm^2		
19	Comprehensive question: First, round 3,856,200 to the 100,000th place to get A, and then use A as the length of the rectangle (width = 100 m). Estimate the area of this rectangle and read the answer.		
20	A square park with sides approximately 200 m. (a) Estimate the area to the nearest hundred thousand (b) If 4 trees are planted in 1 m^2, approximately how many trees can be planted? (c) Round the number of trees to the 10,000s		
21	A square is approximately 180 m long and 120 m wide. (a) Estimate the area (b) Round the answer to the nearest tens of thousands (c) If the cost of the square is $250 per m^2, estimate the total cost (round the answer to the nearest million)		

Knowledge point 4: Comprehensive application questions (subject to sub-test text questions) 🔴 SSPA compulsory exam

Four steps to solve the problem:① **Circle the key numbers and units**(pay attention to m or cm) ②**Judgment operation**(Estimation? Area? Conversion?) ③**Calculate step by step**(Write each step clearly) ④**Write a complete answer**+ Inspection unit

comprehensiveapplicationquestion (total 8 questions, all must do)

#	Question	Difficulty	Working Space
22	The school hall is 35 m long and 22 m wide. If tiles are to be laid, 9 tiles are needed per m^2. How many tiles are needed in total? (Calculate the area first)		
23	A swimming pool is 50 m long and 25 m wide. There is a 2 m wide walkway around the pool. What is the area of the sidewalk in m^2? (Tip: large rectangle − swimming pool)		
24	A theme park in Hong Kong covers an area of approximately 1,260,000 m^2. (a) Write how this number is read (b) Round it to the nearest hundred thousand (c) If there are 50,000 visitors per day, use an approximation to estimate the average m^2 available per person? (Answer is rounded to an integer)		

25	A rectangular farmland is 320 m long and 180 m wide. Express the area in "hectares" (ha) (1 ha = 10,000 m^2), and round the answer to one decimal place.	🌿	
26	Express the area of a rectangular garden (8 m long and 6 m wide) in m^2 and cm^2 respectively. Comparing two numbers, why is the number cm^2 so much larger?	🌿	
27	A large shopping mall has an area of 850,000,000 cm^2. (a) How many m^2 is it equal to? (b) If a shopping mall has 5 floors and each floor has the same area, what is the area in m^2 of each floor?	🌳	
28	The areas of the three parking lots are: A 1,250,000 m^2, B 980,000 m^2, C 1,580,000 m^2. (a) Round each to the 100,000th place (b) Use approximations to find the maximum and minimum area difference (c) Read the answer to (b)	🌳	
29	A rectangular site is approximately 178 m long and 122 m wide. (a) The estimated area is rounded to the nearest ten thousand (b) If one building is built every 100 m^2, approximately how many buildings can be built? (calculated using approximate values)	🌳	

IV. ⛰️ Ultimate challenge area (total 3 questions, 🚀 choose do, SSPA finale level)

#	Question	Difficulty	Working Space
⛰️ 1	A city park covers an area of approximately 3,800,000 m^2. (a) Read this number (b) Convert to km^2 (1 km^2 = 1,000,000 m^2) (c) If the park receives visitors every day, and each person uses an average of 20 m^2 of space, how many people can it receive at most per day? (rounded to the nearest ten thousand)	⛰️	
⛰️ 2	Round 2,468,135 to the 100,000th place to get A, and round to the 10,000th place to get B. (a) How different are A and B? (b) Use A as the side length of the square (unit: m), area = ? m^2 (c) Use B as the length of the rectangle (width = 50 m), area = ? m^2	⛰️	
⛰️ 3	A rectangular site approximately 156 m long and 98 m wide. (a) Estimate the area to the nearest thousand (b) Using the approximation of (a), if one stall is provided for every 25 m^2 of the venue, approximately how many stalls can be installed? (c) If the rent for each stall is $380, what is the approximate total rent? (Take to the nearest 100,000 digits)	⛰️	

V. Class afterhomework

Basic must-doquestions (total 5 questions, mustwrite outcalculateprocess)

#	Question	Difficulty	Working Space
H1	Estimate: A rectangular court approximately 40 m long and 25 m wide. Area approx = ? m^2	🌱	
H2	8 m^2 = ______ cm^2. Read the answer.	🌱	
H3	Read 2,500,000 and write the place value of "2".	🌱	
H4	A playground is 60 m long and 35 m wide. (a) Area = ? m^2 (b) Convert to cm^2	🌱	
H5	Take 5,678,900 to the "100,000" and "10,000" digits.	🌱	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	A park is approximately 150 m long and 120 m wide. The area is estimated and expressed in "10,000 m^2".	🌿	
H7	The rectangular square is 85 m long and 65 m wide. (a) Area = ? m^2 (b) If 4 bricks are laid per m^2, how many bricks are needed? (c) Count the number of bricks to the 10,000s	🌳	
H8	Comparison: Site A has an area of 2,400,000 cm^2 and Site B has an area of 250 m^2, which one is larger? How many m^2 is the difference? (Tip: Unify the units first)	🌳	

VI. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🌱basic questions independently

☐ I can challenge 🌿advanced questions

☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge 🌿Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**m^2 ↔ cm^2 Multiplication error |||SEP|||: 1 m^2 = 100 cm^2 (wrong!)**: 1 m^2 = 100 cm^2 (wrong!)	1 m = 100 cm → 1 m^2 = 100 × 100 = **10,000 cm^2**
2	**I forgot to write "approximately" and the unit when making an estimate.**	The estimated answer must be written as "about" + unit (m^2/cm^2)
3	**Bit value confusion when reading and writing large numbers |||SEP|||: one hundred thousand vs ten thousand vs one million**: One hundred thousand vs ten thousand vs one million	From the right: one hundred thousand/ten thousand hundred thousand million. group of four
4	**After taking the approximate value, use the approximate number to calculate, and forget to mark "≈"**	Each step of calculation using approximate values must be marked "≈", and no points will be deducted when submitting the test.
5	**Wrong use of area formula |||SEP|||: Use "length + width" instead of "length × width"**: Replace "length × width" with "length + width"	Area = length × width (multiply, not add!). Perimeter is plus
6	**Confusing different units in comprehensive questions |||SEP|||: mixed operations of m^2 and cm^2**: m^2 and cm^2 mixed operations	Unify the units before calculating! Convert all to m^2 or all to cm^2
7	**Missed sentences/missed units in application questions**	Must write "Answer:..." + unit, otherwise step points will be deducted